

TRACE-EVC: Text-Guided Relative Affective Control for Zero-Shot Emotional Voice Conversion

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Abstract—Traditional emotional voice conversion (EVC) conditions generation on explicit target emotions like labels or references, defining the target affective state but omitting the direction or nature of the transition. We introduce instruction-guided relative emotional voice conversion, a task where natural-language instructions specify source-conditioned affective transformations (e.g., “make the speech slightly calmer” or “sound noticeably more confident”) instead of fixed targets. To support this task, we construct TRACE-Instruct, a dataset of relative emotion instructions covering categorical transitions, intensity modifications, and open-ended affective changes. We propose TRACE-EVC, a zero-shot framework built around EmoCompass, a module that models each conversion as a source-anchored rectified flow. Rather than conditioning on an explicit target, it predicts the direction and degree of the affective change. Experiments demonstrate that TRACE-EVC accurately follows relative emotion instructions while preserving speaker identity, linguistic content, and speech quality, and remains competitive with conventional EVC systems on standard categorical emotion conversion.

Index Terms—Emotional Voice Conversion, Relative Emotion Transition, Instruction-Guided Control, Valence-Arousal-Dominance

I. INTRODUCTION

Emotional voice conversion (EVC) aims to modify the emotional expression of speech while preserving speaker identity and linguistic content [1], [2]. As an important direction in expressive speech generation, EVC has broad applications in personalized speech interaction, audiobook production, film dubbing, and assistive communication [3], [4]. A central challenge in EVC is how to specify the desired emotional transformation. Existing methods address this through label-based control [5]–[9], reference-based control [10]–[14], and prompt-based control [15]–[17].

Although these control paradigms have been widely studied, they all require an explicit target condition, such as a target emotion category, a reference utterance that demonstrates the desired emotion, or a description of the final target style. In many practical scenarios, however, such target conditions may be unavailable or difficult to specify in a categorical manner. Instead, an alternative, and arguably more natural specification could be expressed as a relative and directional modification of the source utterance: for example, “making the speech sound more confident”, “slightly less excited”, “warmer”, “calmer”, or “more expressive”, while preserving

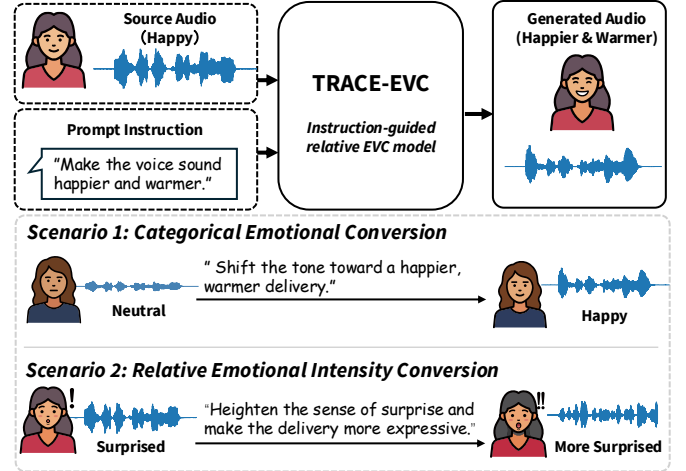


Fig. 1. Illustration of the instruction-guided relative EVC task. The instruction specifies how the source emotion should change rather than a fixed target.

speaker identity and linguistic content. Motivated by this observation, we introduce a new task: instruction-guided relative EVC, in which a natural-language instruction specifies how the emotional expression of a source utterance should be modified, as illustrated in Fig. 1. Unlike conventional EVC settings that convert speech toward a predefined target, this task centers on emotional changes defined relative to the source utterance. Such changes range from categorical emotion shifts to fine-grained modifications of emotional intensity and expressiveness.

This new task introduces two key challenges. First, relative instructions describe affective transformations rather than target emotional states. Such transformations may involve changes in valence, arousal, dominance, or prosody instead of a simple emotion-category switch, requiring an interpretable affective representation that captures both the direction and magnitude of emotional change. Second, existing instruction-style speech datasets describe individual utterances [18]–[20], whereas instruction-guided relative EVC requires descriptions of affective transformations between source and target utterances, leaving a data gap for learning relative emotion instructions.

To address this data gap, we construct TRACE-Instruct, a dataset of relative emotion instructions built from paired emotional speech. Each instruction is generated by prompting

a large language model with the source and target style descriptions together with their affective difference, guiding it to describe the affective transformation rather than simply the target style. The resulting instructions provide free-form descriptions of emotional change without explicitly revealing the target emotion. Covering cross-emotion shifts, intra-emotion intensity modifications, and open-ended affective transitions, TRACE-Instruct provides the training and evaluation data needed for instruction-guided relative EVC.

To address these challenges, we propose TRACE-EVC, a zero-shot framework for instruction-guided relative EVC. Its key component, **Emo-Compass**, rethinks how prompt-based EVC generates emotion embeddings. Prior work such as PromptEVC [15] generates the emotion embedding from a Gaussian noise prior via a diffusion mapper conditioned on the prompt. In our relative setting, however, the source emotion is already given and the instruction is defined relative to it, so we instead anchor the generative process at the source embedding rather than at noise, realizing each conversion as a source-anchored rectified flow whose velocity is exactly the displacement the instruction describes. A synthesis module then generates the converted speech from the source content, speaker embedding, and predicted emotion embedding using established disentanglement techniques [21]–[23], enabling zero-shot conversion without target labels or reference utterances. Experiments demonstrate that TRACE-EVC accurately follows relative emotion instructions while preserving speech quality, speaker similarity, and linguistic content under both cross-emotion and intensity-modification settings.

Our main contributions are summarized as follows:

- **Instruction-guided relative EVC.** We introduce a new EVC task in which natural-language instructions specify source-conditioned affective transformations rather than explicit target emotions, enabling both categorical emotion transitions and fine-grained intensity modification.
- **TRACE-Instruct.** We construct TRACE-Instruct, a dataset of relative emotion instructions that supports training and evaluation for instruction-guided relative EVC across categorical transitions, intensity modifications, and open-ended affective transformations.
- **TRACE-EVC.** We propose TRACE-EVC, whose Emo-Compass casts relative emotion control as a source-anchored rectified flow, enabling zero-shot EVC without target labels or reference utterances while supporting flexible instruction following and fine-grained emotion control.
- **Experimental validation.** Extensive experiments demonstrate that TRACE-EVC follows relative emotion instructions while preserving speech quality, speaker identity, and linguistic content, and remains competitive with conventional EVC systems on standard categorical emotion conversion.

We release the TRACE-Instruct dataset, audio demos, and model code for reproducibility¹.

¹TraceEVC: <https://traceevc4-star.github.io/website/>

II. RELATED WORK

A. Emotional Voice Conversion

Existing EVC methods can be broadly grouped by how the target emotion is specified. Label-based EVC conditions generation on a predefined target emotion category. Early systems transform spectral and prosodic features using statistical or neural models [24], [25], followed by non-parallel adversarial approaches that relax the parallel-data requirement [26], [27]. Subsequent work improves conversion quality and disentanglement [5]–[9], [28] and extends EVC toward speaker-independent and zero-shot settings [21], [22], [29]–[31].

Reference-based EVC instead specifies the desired emotion through a reference utterance. ZEST [10] performs zero-shot audio-to-audio emotion transfer via speaker disentanglement, while TextlessEVC [11], DISSC [12], and HybridVC [14] use decomposed, discrete, or multimodal representations for flexible style transfer.

Prompt-based EVC replaces labels and references with natural-language descriptions of the desired style, offering a more flexible interface. PromptEVC [15] generates a fine-grained emotion embedding from a noise prior via a diffusion-based prompt mapper conditioned on the style prompt, while ClapFM-EVC [16] and PromptVC [17] use natural-language or CLAP-based descriptions to control emotional or stylistic attributes.

Despite these advances, these paradigms remain target-oriented, relying on target emotion labels, reference emotional speech, or target-style descriptions to guide generation. In particular, even prompt-based EVC describes an absolute target style rather than a change from the source. In contrast, our work focuses on source-conditioned emotion transformation, where free-form natural-language instructions describe the desired emotional change relative to the source utterance without requiring target emotion labels or reference speech.

B. Natural-Language-Controlled Speech Generation

Recent work controls speech generation with natural language. Datasets such as TextrolSpeech [18], PromptTTS [32], InstructTTS [33], and SpeechCraft [19] pair utterances with descriptions of their speaking style, emotion, or acoustic attributes, and prompt-based models built on such data condition synthesis on these descriptions to control output style [20], [34]–[38]. In all cases, however, a description characterizes a single utterance, capturing what it should sound like rather than how it should change. In contrast, our instructions describe the emotional change from a source utterance to a target, providing the supervision for relative control.

C. Continuous and Fine-Grained Emotion Modeling

Emotional expression is diverse and cannot be fully captured by a fixed set of emotion labels [39], [40]. To model finer affective variations, prior work explores emotion intensity [3], [7], mixed emotions [4], and dimensional PAD/VAD representations [41]. Recent emotional TTS further explores continuous and intensity-aware control [42]–[48]. Most related to

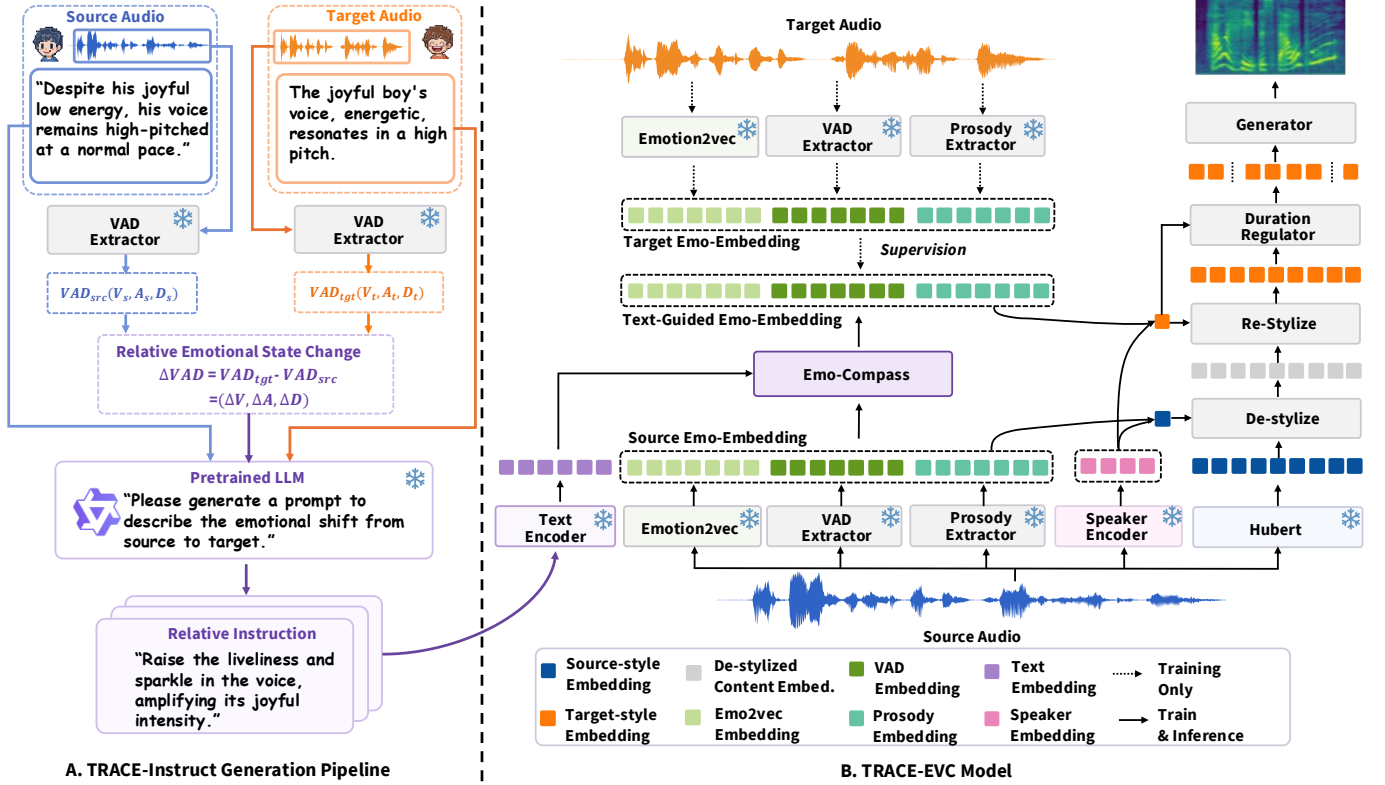


Fig. 2. Overview of the proposed framework: a TRACE-Instruct generation pipeline that produces relative instructions, and a TRACE-EVC model that performs the conversion. The target audio supervises Emo-Compass only during training, shown as dashed training-only paths; at inference, the target emotion embedding is predicted from the source and instruction alone, without any target audio.

ours, EmoSphere-TTS and EmoSphere++ [49], [50] represent emotion as a continuous spherical VAD vector for style and intensity control, but still require a target emotion or intensity to be specified.

Although these studies improve fine-grained controllability, they require an explicit control signal to be specified at inference time, such as an intensity label, a mixture weight, or a target VAD coordinate. In contrast, our work grounds a free-form instruction into a continuous VAD space, where VAD serves as an internal representation of the relative change rather than a control value the user must provide.

III. PROPOSED FRAMEWORK

The proposed framework consists of two components, as illustrated in Fig. 2: TRACE-Instruct, which constructs relative emotion instructions from paired emotional speech, and TRACE-EVC, which performs instruction-guided EVC. Given a source utterance and a relative emotion instruction, TRACE-EVC predicts the target emotion embedding corresponding to the instructed transformation and synthesizes the converted speech.

A. TRACE-Instruct Dataset

We construct TRACE-Instruct, an instruction dataset for relative EVC, as shown in Fig. 2(A). TRACE-Instruct is built from source-target emotional speech pairs, where each pair provides supervision for generating a natural-language instruction describing the relative emotional change from the source toward the target.

Emotional change pair construction. We build source–target speech pairs from two emotional corpora. From the parallel ESD corpus [1], we pair recordings of the same speaker and sentence across different emotions to form inter-emotion pairs. From MEAD [51], we form intra-emotion pairs across intensity levels of the same emotion, and open-ended pairs across different emotions at random intensities. Style descriptions for both corpora are taken from TextrolSpeech [18], denoted p_{src} and p_{tgt} .

Relative affective difference. For each pair, a speech emotion recognition (SER) model [52] predicts the VAD coordinates of the source and target utterances:

$$\begin{aligned} \mathbf{VAD}_{src} &= (V_{src}, A_{src}, D_{src}), \\ \mathbf{VAD}_{tgt} &= (V_{tgt}, A_{tgt}, D_{tgt}). \end{aligned} \quad (1)$$

The relative affective change is then computed as:

$$\Delta \mathbf{VAD} = \mathbf{VAD}_{tgt} - \mathbf{VAD}_{src} = (\Delta V, \Delta A, \Delta D), \quad (2)$$

which captures both the direction and magnitude of the emotional transition from the source toward the target. This relative difference provides the LLM with explicit information about how the affective state should change, guiding it to describe the transformation rather than merely the target style.

Instruction generation. Given the source style description p_{src} , the target style description p_{tgt} , and the relative affective change $\Delta \mathbf{VAD}$, we prompt Qwen3 [53] to generate a set $\mathcal{P}$ of lexically diverse relative emotion instructions:

$$\mathcal{P} = \text{LLM}(p_{src}, p_{tgt}, \Delta \mathbf{VAD}). \quad (3)$$

To keep the instructions relative without revealing the target, we apply rule-based filtering that removes explicit target emotion labels, intensity levels, numerical VAD values, and dataset metadata.

B. TRACE-EVC

TRACE-EVC consists of two cascaded modules: **Emo-Compass** and a **Synthesis Module**, as shown in Fig. 2(B). Given the source emotion embedding and a natural-language instruction, Emo-Compass predicts a target emotion embedding, which the Synthesis Module then uses, together with the source content and speaker representations, to generate the converted speech. Because the speaker is inherited from the source and emotion is conditioned on a continuous embedding rather than a discrete label, TRACE-EVC supports zero-shot conversion of unseen speakers without target labels or reference utterances at inference time.

Emo-Compass. Emo-Compass predicts the target emotion embedding $\hat{\mathbf{z}}_{tgt}$ from the source emotion embedding $\mathbf{z}_{src}$ and the instruction embedding $\mathbf{p}$, obtained from a frozen text encoder [54]. Each emotion embedding concatenates an emotion2vec [55] embedding $\mathbf{e}$, VAD coordinates $\mathbf{a}$, and mean prosodic features $\mathbf{r}$ (mean F_0 and energy), giving $\mathbf{z} = [\mathbf{e}; \mathbf{a}; \mathbf{r}]$. emotion2vec captures category-level emotional information but is less effective at modeling fine-grained intensity variations; we therefore add VAD and prosodic cues to complement it for continuous affective and intensity-related changes. Ablation studies in Sec. IV-C validate this design.

A relative instruction specifies a displacement rather than an absolute endpoint, so the source emotion is a natural starting point. Emo-Compass therefore realizes this source-anchored prediction as a rectified flow in the affective space, whose velocity parameterizes that displacement. Given the source embedding $\mathbf{z}_{src}$ and the target emotion embedding $\mathbf{z}_{tgt}$, we define a linear trajectory between them as

$$\mathbf{x}(s) = (1 - s)\mathbf{z}_{src} + s\mathbf{z}_{tgt}, \quad (4)$$

where s denotes the flow time. A Transformer D_θ , conditioned on the current state $\mathbf{x}(s)$, the flow time s , the source anchor $\mathbf{z}_{src}$, and the instruction embedding $\mathbf{p}$, predicts a velocity $\hat{\mathbf{v}} = D_\theta(\mathbf{x}(s), s; \mathbf{z}_{src}, \mathbf{p})$, trained to match the constant target velocity along the trajectory:

$$\mathbf{v}^* = \mathbf{z}_{tgt} - \mathbf{z}_{src}. \quad (5)$$

This target velocity $\mathbf{v}^*$ is exactly the relative emotional change from the source toward the target, and is what Emo-Compass learns to predict. Because each training pair shares the same speaker, the learned displacement $\mathbf{v}^*$ captures emotion-induced variation rather than speaker identity, reducing speaker-related leakage. During inference, the target is unknown, so we integrate the predicted velocity. Starting from $\mathbf{x}^{(0)} = \mathbf{z}_{src}$, we take K Euler steps:

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \frac{1}{K} D_\theta(\mathbf{x}^{(k)}, \frac{k}{K}; \mathbf{z}_{src}, \mathbf{p}), \quad \hat{\mathbf{z}}_{tgt} = \mathbf{x}^{(K)}. \quad (6)$$

This yields the text-guided target emotion embedding $\hat{\mathbf{z}}_{tgt}$.

The training objective of Emo-Compass combines a velocity-regression term with four auxiliary guidance terms. The velocity-regression term matches the predicted velocity to the ground-truth displacement $\mathbf{v}^*$:

$$\mathcal{L}_{flow} = \mathbb{E}_s [\|D_\theta(\mathbf{x}(s), s; \mathbf{z}_{src}, \mathbf{p}) - \mathbf{v}^*\|^2]. \quad (7)$$

Since this term alone lets the high-dimensional emotion2vec component dominate the VAD and prosody cues (Sec. IV-C), we add four guidance terms on a single-step estimate $\tilde{\mathbf{z}}_{tgt}$ of the target, obtained by extrapolating the predicted velocity $\hat{\mathbf{v}}$ to the endpoint. Two terms constrain the target emotion: a direction term $\mathcal{L}_{cos}$ and a category term $\mathcal{L}_{emo}$. The other two supervise the continuous cues: a reconstruction term $\mathcal{L}_{aff}$ on VAD and prosody, and an intensity term $\mathcal{L}_{int}$ enforcing intensity-consistent ordering (on MEAD). The overall objective is:

$$\mathcal{L} = \mathcal{L}_{flow} + \lambda_{cos}\mathcal{L}_{cos} + \lambda_{emo}\mathcal{L}_{emo} + \lambda_{int}\mathcal{L}_{int} + \lambda_{aff}\mathcal{L}_{aff}. \quad (8)$$

Synthesis Module. The Synthesis Module adapts the DurFlex-EVC [8] backbone, replacing its discrete speaker and emotion labels with continuous embeddings to enable zero-shot conversion. It takes the source content from HuBERT [56], the source speaker embedding $\mathbf{s}$ from EASE [10], and the source and predicted target emotion embeddings $\mathbf{z}_{src}$ and $\hat{\mathbf{z}}_{tgt}$.

The speaker and emotion embeddings are first combined into source and target style representations through an emotion projection ϕ_{emo} :

$$\mathbf{m}_{src} = \mathbf{s} + \phi_{emo}(\mathbf{z}_{src}), \quad \mathbf{m}_{tgt} = \mathbf{s} + \phi_{emo}(\hat{\mathbf{z}}_{tgt}). \quad (9)$$

The two style vectors share the speaker term $\mathbf{s}$ and differ only in emotion: $\mathbf{m}_{src}$ de-stylizes the content and $\mathbf{m}_{tgt}$ re-stylizes it, converting emotion while preserving content and speaker identity.

Following DurFlex-EVC [8], the re-stylized content is processed by a duration regulator and a conditional diffusion decoder. The duration predictor is trained in the log domain:

$$\mathcal{L}_{dur} = \|\log \mathbf{n} - \log \hat{\mathbf{n}}\|_2^2, \quad (10)$$

where $\mathbf{n}$ and $\hat{\mathbf{n}}$ denote the ground-truth and predicted unit durations, respectively. The decoder then generates the mel-spectrogram by denoising from Gaussian noise. With $\mathbf{y}_t$ the mel-spectrogram noised with Gaussian noise ϵ at diffusion step t , and $\mathbf{c}$ the frame-level content condition, the decoder ϵ_θ is trained to predict ϵ :

$$\mathcal{L}_{mel} = \mathbb{E}_{t, \epsilon} [\|\epsilon_\theta(\mathbf{y}_t, t \mid \mathbf{c}, \mathbf{m}_{tgt}) - \epsilon\|_2^2]. \quad (11)$$

A HiFi-GAN [57] vocoder converts the generated mel-spectrogram into waveform.

IV. EXPERIMENTS AND RESULTS

A. Experimental Setup

Datasets. We conduct experiments on ESD [1] and MEAD [51]. For inter-emotion conversion, we use the English portion of ESD, covering five emotions: Neutral, Happy, Angry, Sad, and Surprise, with 17,500 utterances and about 11.9

TABLE I

COMPARISON WITH EVC BASELINES. EECS, SECS, AND UTMOS ARE REPORTED AS DECIMALS; ACC_{cls} AND WER ARE PERCENTAGES. TRAINING HOURS DENOTES THE AMOUNT OF SPEECH USED TO TRAIN EACH SYSTEM. VEVO IS TRAINED ON FAR MORE DATA (101K H) AND IS INCLUDED ONLY AS A LARGE-SCALE REFERENCE, NOT AS A DIRECTLY COMPARABLE BASELINE UNDER OUR 11.9 H SETTING.

Model	Control Input	Training Data(h)	WER↓	EECS↑	ACC_{cls} ↑	SECS↑	UTMOS↑	nMOS↑
SGEVC	Emotion Label	11.9	9.81	0.80	80.44	0.60	3.23	3.84
DurFlex-EVC	Emotion Label	11.9	10.05	0.81	89.50	0.56	3.17	3.67
ZEST	Speech Reference	11.9	8.09	0.79	78.58	0.55	3.16	3.36
VEVO	Speech Reference	101k	8.48	0.52	32.42	0.64	3.61	3.87
TRACE-EVC	Natural Language Prompt	11.9	9.96	0.82	93.00	0.68	3.57	3.98
w/o Emo-Compass	Natural Language Prompt	11.9	10.90	0.65	43.08	0.66	3.38	–

hours. We follow the standard split and hold out 30 sentences per emotion per speaker for testing, using a seen-speaker setting to enable direct comparison with prior EVC baselines. For intra-emotion intensity conversion, we use MEAD, which contains four emotions with three intensity levels. We adopt a speaker-disjoint split with 36 training actors, 3 validation actors, and 4 unseen test actors, evaluating zero-shot conversion on unseen speakers. After retaining paired samples across intensity levels, the MEAD subset contains 12,350 utterances and about 13.8 hours. Both settings provide paired source–target recordings for direct comparison. The seen-speaker ESD setting is adopted only to match the evaluation protocol of prior EVC baselines, while our zero-shot capability is assessed on the unseen-speaker MEAD split.

Baselines. We compare TRACE-EVC with representative EVC baselines under two conventional control paradigms: reference-based control and label-based control. Reference-based baselines include VEVO [13], [58], [59] and ZEST [10], which use a speech reference to guide emotion conversion. Label-based baselines include SGEVC [9] and DurFlex-EVC [8], which use discrete emotion labels as control inputs. The control inputs of all systems are summarized in Table I.

We do not include prompt-based EVC models such as PromptEVC [15], as no public implementation is available. Moreover, their prompts specify absolute target styles rather than relative affective changes, making them not directly comparable to our instruction-guided setting.

Metrics. All objective metrics are averaged over the test set. We measure speech intelligibility using WER computed by Whisper-large-v3 [60]. Emotion similarity and speaker similarity are measured by the cosine similarity between the converted audio and the ground-truth target recording, using utterance-level emotion2vec [55] embeddings (EECS) and Resemblyzer [61] speaker embeddings (SECS), respectively. A higher EECS indicates that the converted speech reaches the intended target emotion, reflecting more accurate emotion conversion. Speech quality is measured by UTMOS [62] on ESD; since UTMOS does not support the 48 kHz audio of MEAD, we use NISQA [63] to assess naturalness on MEAD. To provide an emotion-related evaluation independent of the emotion2vec-based EECS metric, we also report emotion classification accuracy (ACC_{cls}) using a wav2vec2-XLSR [64] classifier fine-tuned on ESD.

Subjective Evaluation Protocol. We recruit 15 participants with normal hearing and proficient English to rate naturalness

(nMOS) and instruction following (IF) on a 1–5 scale. For nMOS, each system is evaluated on the same 8 conversion pairs from the test set. For IF, each setting is evaluated on 10 randomly picked converted utterances. Raters are presented with the source utterance, converted utterance, and instruction, and score how well the converted speech realizes the instructed relative change. This paired presentation encourages raters to focus on the relative change rather than the absolute emotion. Each sample is scored by all 15 raters, and we report the mean.

Implementation Details. The natural language instruction is encoded by a frozen E5 [54] encoder, and content is represented by HuBERT [56] discrete units over a 200-cluster codebook. Emo-Compass is a 6-layer Transformer trained as a source-anchored rectified flow. The two modules are trained separately with AdamW at a learning rate of 1×10^{-4} : Emo-Compass for 100 epochs and the synthesis module for about 40k steps, on 4 A100 GPUs. The vocoder is a HiFi-GAN [57] pretrained on LibriTTS [65].

B. Results on EVC

Inter-emotion. As shown in Table I, TRACE-EVC achieves the highest EECS, ACC_{cls} , and SECS, indicating effective emotion conversion and speaker preservation, while maintaining competitive WER and UTMOS in terms of intelligibility and speech quality. Compared with reference-based methods, TRACE-EVC requires no reference utterance and achieves higher speaker similarity. Compared with label-based methods, it is not limited to discrete emotion labels and supports relative emotion control through natural-language instructions. Removing Emo-Compass and concatenating the source emotion and instruction embeddings instead markedly degrades emotion conversion, underscoring the value of explicitly predicting the target emotion embedding.

Intra-emotion. We evaluate intra-emotion intensity conversion on MEAD under seen- and unseen-speaker conditions. As shown in Table II, TRACE-EVC obtains NISQA scores close to those of the ground-truth recordings under both conditions, indicating that the converted speech retains good naturalness. Speaker and emotion similarities remain stable across both conditions, suggesting that TRACE-EVC preserves speaker identity while producing the target intensity pattern. Note that EECS is computed against each split’s own target and is therefore not directly comparable across conditions. The higher WER relative to the ground-truth reflects the added difficulty of converting expressive, high-arousal MEAD speech,

TABLE II
INTRA-EMOTION CONVERSION RESULTS ON MEAD. SECS AND EECS ARE REPORTED AS DECIMALS AND WER AS A PERCENTAGE.

System	NISQA $\uparrow$	SECS $\uparrow$	EECS $\uparrow$	WER $\downarrow$
Unseen speakers				
GT	3.248	1.0	1.0	8.6
Ours	3.147	0.706	0.914	14.8
w/o Emo-Compass	3.011	0.698	0.902	16.6
Seen speakers				
GT	3.366	1.0	1.0	11.9
Ours	3.193	0.831	0.855	16.5
w/o Emo-Compass	3.052	0.802	0.851	18.4

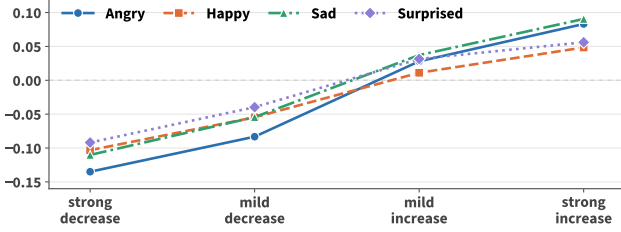


Fig. 3. Mean arousal shift under relative intensity instructions on MEAD. Free-form instructions are grouped by direction and strength, from strong decrease to strong increase. Arousal shifts increase monotonically across all emotions.

on which ASR is already challenging even for real recordings. Replacing Emo-Compass with a direct concatenation of the source emotion and instruction embeddings consistently degrades the metrics, demonstrating the benefit of explicitly predicting an instruction-guided target emotion embedding before synthesis. Fig. 3 shows that the mean arousal shift increases monotonically with the instructed intensity direction and strength across all four emotions, indicating that TRACE-EVC supports graded relative intensity control. We analyze arousal because it is closely associated with emotional activation and intensity among the affective dimensions.

Instruction following. As shown in Table III, TRACE-EVC achieves high IF scores in both inter-emotion and intra-emotion settings, confirming that it follows relative instructions for both categorical conversion and intensity modification.

C. Emo-Compass Evaluation

We evaluate Emo-Compass with four component-level metrics: emotion classification accuracy (emo_acc), EECS between the predicted and ground-truth target emotion embeddings, VAD L2 distance between their VAD coordinates, and intensity MSE (int_MSE). Unlike the end-to-end ACC_{cls} in Table I, emo_acc applies an emotion2vec-based classifier directly to the predicted target emotion embedding $\hat{z}_{tgt}$ rather than to synthesized audio. We also compare with a random permutation baseline, which shuffles the target emotion embeddings across test pairs, preserving the target marginal distribution while breaking the source-instruction-target correspondence.

As shown in Table IV, Emo-Compass substantially outperforms this baseline on both ESD and MEAD, including the unseen-speaker MEAD split. This indicates that Emo-Compass learns instruction-conditioned affective changes rather than re-

TABLE III
INSTRUCTION-FOLLOWING (IF) SCORES, RATED ON A 1–5 SCALE.

	Inter-emotion (ESD)	Intra-emotion (MEAD)
IF $\uparrow$	4.135	4.296

TABLE IV
INTRINSIC EVALUATION AND CUE ABLATION OF EMO-COMPASS. EMO_ACC, EECS, AND VAD L2 ARE REPORTED AS DECIMALS, AND INT_MSE AS A MEAN SQUARED ERROR. RANDOM DENOTES A PERMUTATION BASELINE ON HELD-OUT TEST PAIRS.

System	emo_acc $\uparrow$	EECS $\uparrow$	VAD L2 $\downarrow$	int_MSE $\downarrow$
ESD				
Emo-Compass	0.994	0.949	0.125	–
w/o VAD	0.996	0.953	0.159	–
w/o Prosody	0.995	0.944	0.130	–
w/o VAD & Prosody	0.995	0.952	0.160	–
Random	0.204	0.491	0.315	–
MEAD				
Emo-Compass	0.991	0.965	0.115	7.15
w/o VAD	0.997	0.969	0.191	14.13
w/o Prosody	0.993	0.964	0.121	9.85
w/o VAD & Prosody	0.992	0.966	0.190	14.98
Random	0.244	0.586	0.281	28.79

lying on dataset-level target statistics. The cue ablation shows the complementary roles of the affective cues. emotion2vec provides strong category-level information, with high emo_acc and EECS across all variants. Adding VAD and prosody leaves these category metrics essentially unchanged while markedly improving the continuous metrics: removing VAD increases VAD L2 and int_MSE , and removing prosody further degrades int_MSE on MEAD. Overall, these results support combining emotion2vec, VAD, and prosodic features in Emo-Compass.

D. TRACE-Instruct Analysis

Instruction diversity. As shown in Table V, TRACE-Instruct attains a slightly higher Hypergeometric Distribution D (HD-D) [66] than TextrolSpeech [18], indicating marginally greater lexical diversity. Its low Self-BLEU [67] further suggests limited redundancy among the generated instructions, whereas per-pair Self-BLEU is undefined for single-prompt TextrolSpeech. In terms of instruction length, TRACE-Instruct produces consistently longer and more descriptive instructions. Overall, it offers lexically diverse, low-redundancy descriptions of emotional transitions for instruction-guided emotion control.

TABLE V
INSTRUCTION DIVERSITY. TRACE-INSTRUCT IS AVERAGED OVER ESD/MEAD; PER-PAIR S-BLEU IS N/A (–) FOR SINGLE-PROMPT TEXTROLSPEECH. INSTRUCTION LEN. IS THE WORD COUNT PER INSTRUCTION.

Dataset	#Prompts	HD-D $\uparrow$	S-BLEU $\downarrow$	Instruction Len.	
				Mean	Std
TRACE-Instruct	0.54M	0.776	0.028	16.99	4.67
TextrolSpeech	0.24M	0.774	–	12.64	3.97

V. CONCLUSION

We introduced instruction-guided relative EVC, where natural-language instructions describe emotional changes relative to the source rather than a fixed target, and built

TRACE-Instruct to support it across inter-emotion, intensity, and open-ended changes. Our zero-shot framework TRACE-EVC models each instruction as a source-anchored trajectory in a continuous affective space, following relative instructions while preserving quality, speaker similarity, and intelligibility, and remaining competitive with conventional EVC baselines.

VI. ACKNOWLEDGMENT

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GENERATIVE AI USE DISCLOSURE

Generative AI tools were employed solely for language polishing of text written by the authors. These tools were not used to generate scientific content, results, experimental designs, analyses, or conclusions. All authors are responsible for the full content of this paper and consent to its submission.

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